Status	Finished
Started	Wednesday, 28 October 2026, 10:41 PM
Completed	Wednesday, 28 October 2026, 10:55 PM
Duration	14 mins 14 secs
Grade	**15.00** out of 15.00 (**100**%)

Question **1**

Complete

Mark 1.00 out of 1.00

How does Merge Sort achieve its efficiency?

- a. By comparing elements sequentially
- b. By sorting data in a single pass
- c. By using the bubble-up method
- d. By breaking the input into smaller parts and merging them

The correct answer is: By breaking the input into smaller parts and merging them

Question **2**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after the two halves of the list are sorted?

- a. They are combined to form a single sorted list
- b. They are split again into smaller sublists
- c. They are discarded
- d. They are compared element by element

The correct answer is: They are combined to form a single sorted list

Question **3**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

- a. Quick Sort
- b. Bubble Sort
- c. Insertion Sort
- d. Merge Sort

The correct answer is: Merge Sort

Question **4**

Complete

Mark 1.00 out of 1.00

Two-way merge sort algorithm is used to sort the following elements in ascending order.
200,470,150,80,90,40,400,300,120,70
What is the order of these elements after second pass of the merge sort algorithm?

- a. 200,470,80,150,40,90,300,400,70,120
- b. 40,80,90,150,200,300,400,470,70,120
- c. 80,150,200,470,40,90,300,400,70,120
- d. 40,70,80,90,120,150,200,300,400,470

The correct answer is:
80,150,200,470,40,90,300,400,70,120

Question **5**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

- a. Provides a basis for other algorithms to work efficiently
- b. Decreases the efficiency of algorithms
- c. Makes data harder to manage
- d. Makes code execution slower

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **6**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It allows for quicker identification of duplicates
- b. It has no effect on the analysis process
- c. It complicates the analysis process
- d. It makes the analysis slower

The correct answer is: It allows for quicker identification of duplicates

Question **7**

Complete

Mark 1.00 out of 1.00

What is one of the first steps in a divide-and-conquer algorithm like Merge Sort?

a. Comparing each element with the others
b. Dividing the input into smaller subproblems
c. Combining sorted sublists
d. Sorting the entire list sequentially

The correct answer is: Dividing the input into smaller subproblems

Question **8**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

a. Arranging data in a specific order
b. Finding a specific element in a list
c. Merging two datasets
d. Removing duplicates from a list

The correct answer is: Arranging data in a specific order

Question **9**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Selection
b. All of the above
c. Duplicates
d. Searching

The correct answer is: All of the above

Question **10**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

a. It makes the data unsorted
b. It slows down the process
c. It complicates the selection of items
d. It makes it easier to select items based on their relationship to the rest of the items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **11**

Complete

Mark 1.00 out of 1.00

Which built-in Python function is used to sort data?

a. sort()
b. order()
c. arrange()
d. sorted()

The correct answer is: sorted()

Question **12**

Complete

Mark 1.00 out of 1.00

_____is putting an element in the appropriate place in a sorted list yields a larger sorted order list.

a. Extraction
b. Distribution
c. Selection
d. Insertion

The correct answer is:
Insertion

Question **13**

Complete

Mark 1.00 out of 1.00

Which algorithm typically follows a divide-and-conquer structure?

a. Linear Search
b. Bubble Sort
c. Merge Sort
d. Binary Search

The correct answer is: Merge Sort

Question **14**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

a. Rearranging data without sorting
b. Dividing the input into parts, solving each part, and combining the solutions
c. Sorting data in a single pass
d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **15**

Complete

Mark 1.00 out of 1.00

What is the primary advantage of the divide-and-conquer approach in sorting algorithms?

a. It avoids the need for recursion
b. It only works on small datasets
c. It simplifies the sorting process by using only one pass
d. It allows for efficient parallel processing and sorting of data

The correct answer is: It allows for efficient parallel processing and sorting of data